



UNITED REPUBLIC OF TANZANIA

INFORM Subnational Disaster Risk

The Country Model Template — how risk is calculated, the bottleneck, and scaling it to local level

Prime Minister's Office – Disaster Management Department (PMO-DMD)



INFORM: how the Country Model Template is built

- INFORM is an open composite index of disaster risk — one 0–10 score per area (0 = lowest, 10 = highest). Tanzania runs the regional INFORM Country Model Template (Excel); the national INFORM risk is 4.1.
- **Risk is built from three dimensions:**
 - Hazard & Exposure — drought, flood, earthquake, cyclone, conflict ... and who/what is exposed
 - Vulnerability — poverty, health, nutrition, food insecurity, displacement
 - Lack of Coping Capacity — institutions & infrastructure (DRR, governance, WASH, health, education)
- **Structure: 78 raw indicators (53 used) → 6 categories → 3 dimensions → 1 risk score, for every unit.**



How an indicator becomes a 0–10 score (the hidden-sheet method)

- 1. Actual value, in its natural unit
- 2. Denominator — ÷ population / area / GDP, or none
- 3. Outlier cap — Tukey IQR fence: $\text{MAX}(\text{MIN}(x, Q3 + 1.5 \cdot \text{IQR}), Q1 - 1.5 \cdot \text{IQR})$
- 4. Transform — None | Logarithm $\text{LN}(0.001 + x)$ | Exponential
- 5. Min-max scaling — $10 \times (x - \text{Min}) / (\text{Max} - \text{Min})$ against a fixed reference range
- 6. Direction — if protective (higher = better): $10 - \text{score}$
- **7. Clamp to [0,10] → ROUND to 1 decimal → the 0-10 indicator**
- Every indicator, whatever its unit, passes through the same chain — Tukey outliers, log/exp transform, min-max, direction.



A few indicators — many units (why standardisation is needed)

Indicator	Unit	Dimension
Food security – insufficient food	% of population	Vulnerability
Historic drought frequency	years	Hazard & Exposure
Internally displaced people	count (people)	Vulnerability
Gross National Income per capita	US\$ (thousands, log)	Coping Capacity
Health expenditure per capita	US\$ per capita	Coping Capacity
Physicians density	per 10,000 population	Coping Capacity
Measles incidence	per 1,000,000 population	Vulnerability
Soil erosion	Mg / ha / yr	Hazard & Exposure
Coastal erosion	metres / year	Hazard & Exposure

%, years, counts, US\$, per-1,000, per-10,000, Mg/ha/yr, m/yr — all reconciled onto one 0–10 scale.



From indicators to risk — categorisation & aggregation

- **Component** = AVERAGE of its indicators where Use = "Yes" (Excel AVERAGEIFS)
- **Category** = AVERAGE of its components (arithmetic mean)
- **Dimension** = scaled GEOMEAN of its categories
- **Risk** = $\sqrt[3]{H \times V \times LCC}$ — **cube-root of the three dimensions**
 - $\text{ROUND}(H^{1/3} \times V^{1/3} \times LCC^{1/3}, 1)$
- The geometric mean matters: a unit cannot "average away" one very weak dimension — high hazard with low coping capacity still yields high risk.



The honest problem: it all lives in deeply hidden Excel sheets

- Every reference range, outlier fence, transform and aggregation step is buried in hidden worksheets.
- Opaque — the math is invisible to the people who must use and trust the numbers.
- Fixed unit set — built for ~170 districts; it cannot scale to 195 councils, wards or villages.
- Manual & single-file — one workbook, no version control, easy to break, hard to audit.
- Closed to local data — no clean way to add a local indicator or feed council-level values.
- No automation — no live update, no link to early warning or anticipatory action.
- Result: the model stops at national / regional level. Local risk — where decisions are made — is out of reach.



We rebuilt the hidden sheets as an engine — proven identical, built to scale

- The hidden-sheet formulas were reverse-engineered and reproduced exactly in open code.
- **Proven byte-for-byte against the official workbook:**
 - 8,664 / 8,664 standardised indicator cells reproduce the template exactly
 - 170 / 170 district risk scores reproduce the template exactly
- Now it scales — the same engine runs unchanged on 195 councils and 31 regions.
- Spec-driven & transparent — add, delete or amend an indicator without touching the math; fully testable.
- **The regional INFORM core is never altered — it is reproduced exactly; everything new is additive.**



Capturing real local data — the methods



- Federated collection tools — one shared Excel per sector; they enter ACTUAL values in natural units and a locked formula shows the 0–10 live.
- One accountable stakeholder per indicator — TMA (climate), GST (geology), NEMC (environment), MoH (health), NBS (statistics), MoA/MUCHALI (food), MoW (water), PMO-DMD (DRR).
- Bottom-up aggregation — village → district → region → country; finer, well-filled data refines the higher level.
- Authentic sources — NBS 2022 Census, TMA CHIRPS/ERA5, GST seismic, MoH TDHS, IPC/MUCHALI, TASAF, and satellite Earth Observation (validated before use).





The gap — what still has to be solved

- Subnational risk is not yet operational — decisions happen at council / ward level, but INFORM stops at national / regional.
- Local data is scattered and uneven — many key indicators exist only at region or national level.
- No sustainable pipeline — capturing, validating and updating local data is still manual.
- Earth-Observation indicators (vegetation, soil moisture, flood, landslide) need proper computation, downscaling and validation.
- No live link to action — risk is not yet wired to the emergency operations centre, early warning, or local planning & budgeting.
- Capacity & financing gaps at local-government level — tools, training and incentives to keep the data flowing.



Opportunities — research & financial support for member-state adoption



- **RESEARCH**

- Earth-Observation computation & statistical downscaling to council / ward level
- Validation against ground events; multi-source "exploded" indicators; climate-seasonality into early warning

- **FINANCIAL SUPPORT & ADOPTION**

- Fund the local-data integration and the federated tools; build local-government capacity to capture & submit
- Scale the transparent engine across SADC member states; integrate risk into EOCC, anticipatory action & local planning
- **Tanzania as the subnational-INFORM pilot — a shared, open method and data standard for the region.**



Conclusion



- The Country Model Template math is fully understood and reproduced exactly — 8,664/8,664 cells, 170/170 risks.
- It now runs at council level (195) because it is open code, not a hidden-sheet workbook.
- Local-data capture and Earth Observation are working, with each layer validated before use.
- **Ask: research partnership + financing to integrate local data and support member-state adoption and local-level integration.**

